

1 Question (20 points)

Give CFG's with the specified number of variable recognizing each of the following languages:

- $A = \{w \mid \text{the length of } w \text{ is odd and its middle symbol is a } 1\}$ (1 variable)
- $B = \{w \mid \text{the set of strings over alphabet } \{a, b\} \text{ with more } a\text{'s than } b\text{'s}\}$ (2 variables)

2 Question (20 points)

Is the language $L = \{0^{n^2}1^{n^2} | n \geq 1\}$ context free? Prove or disprove.

3 Question (20 points)

Give the PDA for the following language: $D = \{w\#x \mid w^R \text{ is a substring of } x \text{ for } w, x \in \{0, 1\}^*\}$

4 Question (20 points)

Given the context free language $C = \{a^n b^m c^m d^n e^p f^p \mid n, m, p \geq 0\}$. Try to disprove it by pumping lemma with the string, $a^p b^p c^p d^p e^p f^p$, and give two general example cases where you can successfully pump up, and two general example cases where you can not successfully pump up. In each case, select strings v and y consisting of only one distinct character.

5 Question (20 points)

Give PDA's that generate the following languages.

- $A = \{w \mid \text{the length of } w \text{ is odd and its middle symbol is the same as its first and last symbol}\}$, The alphabet is $\{0, 1\}$.
- $A = \{a^n b^m c^m d^n e^p f^p \mid n, m, p \geq 0\}$